

First name:

Last name:

Student ID:

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Problem 1. Find $\text{curl}(\vec{v})$ with respect to right-handed coordinates x, y, z for

$$\vec{v} = \begin{bmatrix} 2y^2 \\ 5x \\ 0 \end{bmatrix}.$$

[5 marks]

Solution.

$$\begin{aligned} \vec{\nabla} \times \vec{v} &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2y^2 & 5x & 0 \end{vmatrix} = \hat{i} \left(\frac{\partial}{\partial y} 0 - \frac{\partial}{\partial z} (5x) \right) \\ &\quad - \hat{j} \left(\frac{\partial}{\partial x} 0 - \frac{\partial}{\partial z} (2y^2) \right) \\ &\quad + \hat{k} \left(\frac{\partial}{\partial x} (5x) - \frac{\partial}{\partial y} (2y^2) \right) \\ &= \hat{k} (5 - 4y) = \begin{bmatrix} 0 \\ 0 \\ 5 - 4y \end{bmatrix} \end{aligned}$$

What I am expecting:

- definition of $\text{curl} \vec{v}$ (implied or stated)
- correct calculation of cross-product
- correct calculation of derivatives

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Problem 2. Find a line integral of the form $\int_a^b f(\vec{r}(t)) dt$ where

$$f = xyz, \quad C : \vec{r} = [4t, 3t^2, 12t], \quad -2 \leq t \leq 2.$$

[5 marks]

Solution. In this form:

$$\begin{aligned} \int_a^b f(\vec{r}(t)) dt &= \int_{-2}^2 x(t) \cdot y(t) \cdot z(t) dt \\ &= \int_{-2}^2 (4t)(3t^2)(12t) dt \\ &= \int_{-2}^2 144t^4 dt \\ &= \left. \frac{144}{5} t^5 \right|_{-2}^2 \\ &= \frac{144}{5} (2^5 - (-2)^5) \\ &= \frac{144}{5} (32 - (-32)) \\ &= \frac{144}{5} (64) = \frac{9216}{5} \end{aligned}$$

What I am expecting:

- setting up the integral with the right integrand
- correct calculation

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Problem 3. Compute a line integral of the form $\int \vec{F} \cdot d\vec{r}$, where

$$\vec{F} = [xz, yz, xy], \quad C : \vec{r} = [t, t, e^t], \quad 0 \leq t \leq 5.$$

Hint: Integration by parts formula $\int u dv = uv - \int v du$.

[5 marks]

Solution.

$$\vec{r} = [t, t, e^t]$$

$$\Rightarrow \vec{r}' = [1, 1, e^t]$$

Then

$$\int_C \vec{F} \cdot d\vec{r} = \int_0^5 \begin{bmatrix} x(t)z(t) \\ y(t)z(t) \\ x(t)y(t) \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \\ e^t \end{bmatrix} dt$$

$$= \int_0^5 \begin{bmatrix} te^t \\ te^t \\ t^2 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \\ e^t \end{bmatrix} dt$$

$$= \int_0^5 (te^t + te^t + t^2 e^t) dt$$

$$= \int_0^5 2te^t dt + \int_0^5 t^2 e^t dt$$

$$= 2 \int_0^5 te^t dt + \left[t^2 e^t - \int (2t) e^t dt \right]_0^5$$

$$= 2 \int_0^5 te^t dt + [t^2 e^t]_0^5 - 2 \int_0^5 te^t dt$$

$$= 5^2 e^5 - 0 = 25e^5$$

What I am expecting:

- Using $\vec{F} \cdot \vec{r}'$
- getting correct integrand
- using integration by parts correctly
- getting correct final answer.

□

MATH73235

Spring 2025

Quiz 3

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Extra Page: Must submit even if unused.